



A look at ultra wideband, a wireless technology that's faster than Bluetooth and WiFi, has the potential to transmit huge amounts of data very rapidly and can pass through walls, floors and even steel

The average Joe has a small home but a very big appetite for entertainment. All of us would want to zap video images from digital camcorders to our hard drives or personal video recorders without tripping over the cables that connect them. As home entertainment systems become more sophisticated, you will have flat-panel video displays. You would want to hang them like pictures on your wall. Who would then want connector wires to dangle from those beauties?

You must be thinking that we already have Bluetooth and 802.11 wireless technologies to do this. They do a decent job of linking our computers and digital gadgets in the home and office. However, more bandwidth and speed and a low cost is always welcome. It's here that another wireless technology—ultra wideband or UWB—is making waves.

UWB can handle more bandwidth-intensive applications—such as streaming video—than either 802.11 or Bluetooth because it can transmit data about 10 times faster than the typical DSL line, cable modem or 802.11b. It has a data rate

of roughly 100 Mbps, with speeds up to 500 Mbps. Compare that with the maximum speeds of 11 Mbps for 802.11b, called WiFi, which is the technology currently used in most wireless LANs; and 54 Mbps for 802.11a and 802.11g (although both exist on different spectrums). Bluetooth has a data rate of about 1 Mbps.

UWB gives us these extremely high data rates at a lower cost and lower levels of power consumption (making it ideally suited for handhelds and mobiles). But due to Federal Communications Commission (FCC) restrictions, UWB will be used in the home or office over a short distance—up to about 30 feet (10 metres)—and operate at rates of around 100 Mbps. Most UWB devices should operate between 3.1 and 10.6 GHz.

There has been a lot of opposition to UWB on grounds that it would interfere with navigational-guidance systems, radar and satellite transmissions. Big mobile-phone companies even warned the FCC

that UWB transmissions would block telephone calls. Companies developing UWB networking technology maintain that UWB emissions are too low-powered to interfere with other radio signals. It took the FCC three years to approve the use of the technology for commercial wireless networking applications.

At one point, regulators feared that mobile gadgets would interfere with sensitive communications at airports or military installations. What would stop consumers using UWB technology in MP3 players from swapping thousands of music tracks in seconds instead of minutes or even hours?

Eric Boll, senior applications engineer, XtremeSpectrum, explains: "Actually, the FCC didn't limit the distance, but limited the power spectral density in the rules approved in February 2002 for commercial use. Under the FCC rules, 10 meters is about the distance that today's UWB technology can operate while staying within the spectral mask."

Info The wireless portion of the home networking market is forecast to be \$2.5 billion in 2004, growing to \$3.7 billion in 2006.

// We anticipate commercial applications emerging in the consumer market by Christmas 2003. The first products will be display-centric in nature, such as large screen displays, DVD players, DVRs and set-top boxes that have a wireless capability. In 2004, you will have consumer electronics products with embedded UWB capability //

—Eric Boll, senior applications engineer, XtremeSpectrum